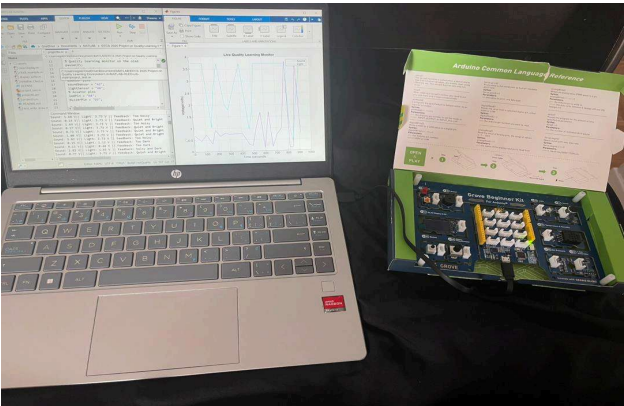


INTRODUCTION

As observed by the United Nations, quality education has only been achieved in a few countries worldwide with many education systems still struggling to meet the expected standards (United Nations, 2024). The aim of my project is to detect light and sound levels that may negatively impact learning within a given environment. The system will provide feedback on the current state of the environment based on these measurements, encouraging those present to maintain a setting that supports effective learning. In doing so, my project aligns with the fourth goal of the United Nations Sustainable Development Goals, which focuses on ensuring quality education for all (United Nations, 2024). The concept of a “quiet learning environment” was highlighted in the project description , focusing primarily on noise detection (Project2025, 2025). I decided to expand this idea by incorporating light detection, as I believe an effective learning environment should not only be quiet but also well-lit. Creating an optimal learning environment is crucial because excessive noise and poor lighting can reduce concentration and negatively impact student performance. By monitoring these factors the system not only supports individual learning but also contributes to broader societal goals by helping ensure equitable access to quality education. My project ultimately aims to create an environment that supports effective learning, thereby advancing the provision of quality education through both technological innovation and practical application.

SYSTEM OVERVIEW



This project was implemented using the Grove Beginner Kit For Arduino, which features a range of pre-soldered sensors and actuators connected directly to the main board. This modular design ensures reliable signal communication and eliminates the need for external wiring. The system is powered and programmed through a laptop connected via Micro USB.

The bulleted list of the Components used is as follows;

Component	Roles	Connections
• Sound sensor	Detects sound intensity in the environment	Analog Port A2
• Light sensor	Detects light intensity in the environment	Analog Port A6
• LED	Emits red light as a visual indicator	Digital Port D4
• Buzzer	Emits sound alerts based on environment state	Digital Port D5
• OLED Display	Provides feedback and displays environment status	I2C Interface

- | | | |
|----------------------------------|--|--|
| • Micro USB Cable | Connects the laptop to the board | Micro USB on the board to the USB port on the laptop |
| • Laptop | Uploads code and powers the system | Micro USB Cable |
| • Grove Beginner Kit for Arduino | Hosts and connects all sensors and actuators | Integrated |

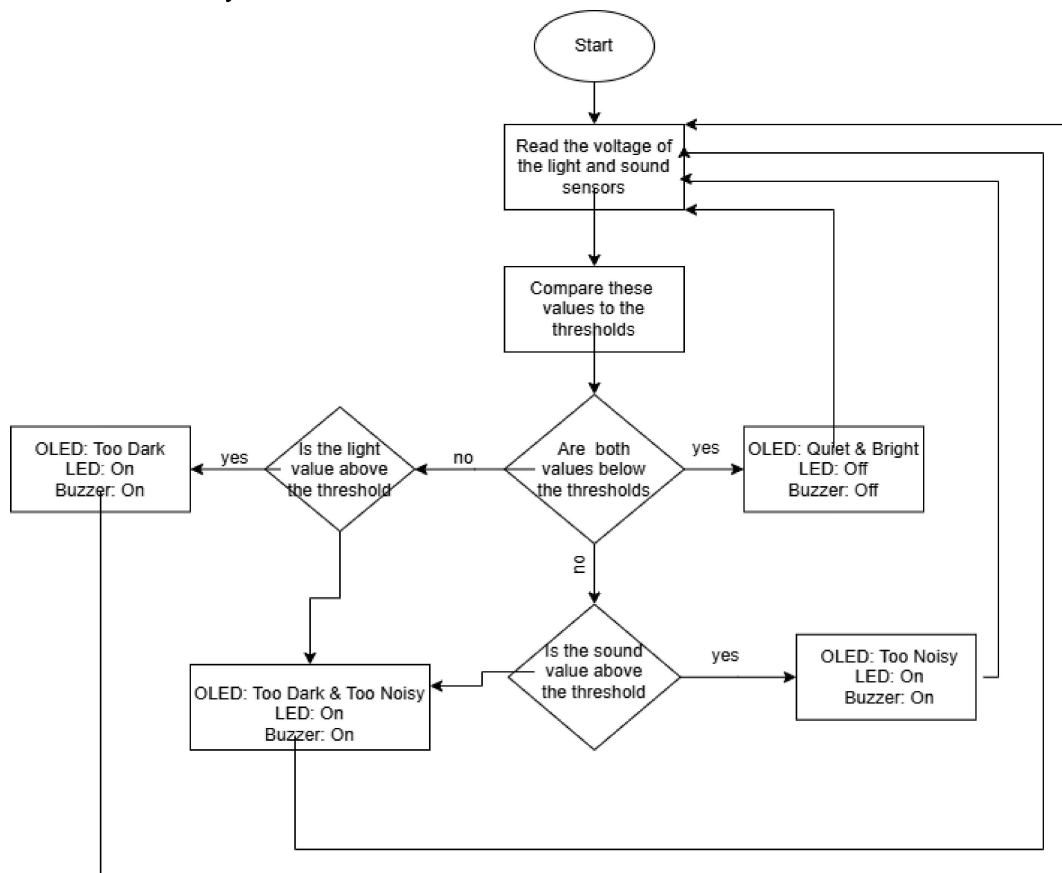
SYSTEM DESIGN

The system monitors environmental conditions by measuring sound and light levels using the sensor built into the arduino board. Sensor values are compared to experimentally determined thresholds to classify the current learning environment and trigger appropriate feedback through the buzzer, LED, and OLED display. The system can indicate four states;

- Quiet and Bright: Optimal conditions; no alerts
- Noisy: Buzzer and LED activate; OLED displays "TOO NOISY".
- Dark: LED and Buzzer activate; OLED displays "TOO DARK".
- Noisy and Dark: LED and Buzzer activate; OLED displays "TOO NOISY AND TOO DARK".

The system runs continuously, providing real-time updates without additional signals filtering, as built-in sensor performance is sufficient. A custom OLED library by Aradhya Chawla enables clear text feedback so users can easily interpret the current environment at a glance.

Flowchart of the system

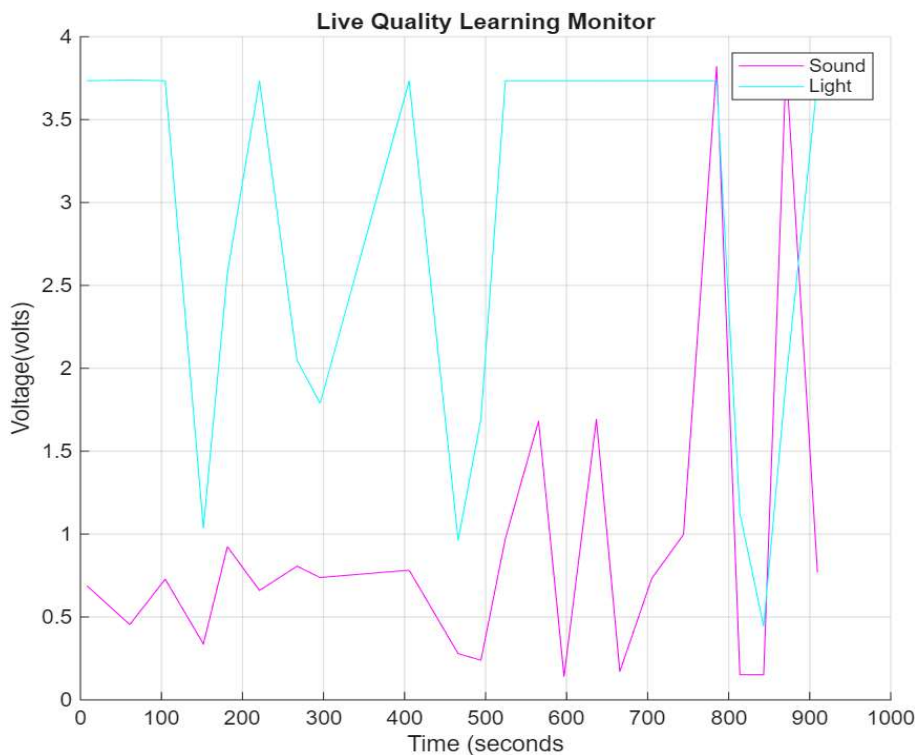


RESULTS

To ensure the system responded effectively to environmental changes, thresholds for light and sound intensity were determined experimentally through trial and error. A light threshold of 2-5 volts was chosen, as readings below this value made doing activities like reading and writing in the environment difficult; and a sound threshold of 1.5 volts was selected, as sound levels above this value became distracting and could reduce the effectiveness of learning.

With these thresholds, the actuators responded accurately: the LED and buzzer activated appropriately when environmental conditions exceeded the set limits. The OLED displayed clear, state-specific messages corresponding to the four environmental conditions: “Quiet and Bright”, “Too Noisy”, “Too Dark”, and “Too Dark and Too Noisy”.

Testing showed that the system correctly detected noise spikes and changes in light intensity. However, it occasionally reacted slightly slowly to sudden changes in both light and sound. This behavior is consistent with the rapid fluctuations in sensor readings, which are not filtered in the current implementation.



The graph above shows the recorded light and sound sensor values over time during testing. Whenever the sensor values exceeded these thresholds, the LED and buzzer were activated, and the corresponding message appeared on the OLED display. The graph demonstrates that the system reliably detected noise spikes and low_light conditions, although occasional small delays in response to sudden changes were observed.

CHALLENGES

During the course of this project, I encountered two major challenges: using the OLED display and writing unit tests for my system. For the OLED display, I used a library created by Aradhya Chawla, which I discovered through a video by Professor James Smith (Smith,2024). At first, learning the syntax of the `display_write` function in the oled library was difficult, and getting the correct text format on the screen required several attempts. However, the documentation provided by the library's author (Chawla,2024) helped me better understand the function, and I was eventually able to successfully display all necessary feedback messages to the user.

Writing unit tests was another challenge, as I encountered a lot of errors in implementing these tests. With the help from MATLAB's official Youtube channel (MathWorks, 2016), I learned how to isolate and evaluate the core decision-making code.

Overall, these challenges taught me the importance of using documentation and external learning resources. If more time were available, I would improve the system by adding sound filtering to make the system suitable for classrooms, where the teacher's voice should not trigger the noise alert.

USE OF AI

Artificial Intelligence tools played a supporting role in this project, mainly for coding clarity and documentation. MATLAB Copilot's auto-fill feature was helpful in arranging code neatly and improving readability. ChatGPT was also used to refine grammar, structure the report and ensure proper reference formatting, which made the written communication more professional.

However, when attempting to use AI for the OLED display code, the results were not very successful at first. Before reading the actual documentation provided by the OLED library's author, I asked ChatGPT for help, but the responses did not align well with the pattern of code my program used and occasionally introduced incorrect syntax. For this part of the project, AI solutions were either not fully functional or too different from the expected approach, so I relied instead on the official documentation and external tutorial videos.

This experience showed that while AI can assist with understanding, formatting, and improving writing quality, it does not always replace the need for direct technical resources or hands-on experimentation. In the end, AI supported my learning without replacing it, and I gained a clearer sense of when AI is helpful and when traditional engineering problem-solving is still the best approach.

Reference list

- 1.) United Nations. (2024). Sustainable Development Goal 4: Quality Education. <https://sdgs.un.org/goals/goal4>.
- 2.) Project Description Document. (2025). 2.1 Topics 4: Quiet Learning Environment Monitor. Retrieved from [Project2025.pdf](#)

- 3.) Chawla, A. (2023). OLED library documentation. GitHub repository.
<https://github.com/AradhyaC/MATLAB-OLED-Lib>.
- 4.) MathWorks. (2024). Unit testing with MATLAB coder. Youtube.
<https://www.youtube.com/watch?v=BeoaB6bZuZA&list=LL&index=1>
- 5.) Smith, J.(2024). Grove OLED tutorial. Youtube.
<https://www.youtube.com/watch?v=t22pSEy9weQ>